LAYER 3 · NETWORK

L3

Addresses, Subnets and Routes

Getting a packet from one network to another: IPv4 and IPv6 addresses, subnetting, static and OSPF routing, ACLs, NAT and the two tools that test it all.

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IPv4 Addressing

What the four numbers mean, how the mask splits network from host, and the special ranges.



The mask's 255s mark the network part. Whatever is left over numbers the hosts.

An IPv4 address is 32 bits, written as four numbers from 0 to 255 separated by dots: 192.168.10.25. On its own an address is half the story. The **subnet mask** says which part names the network and which part names the device on it.

Network part and host part

With the mask 255.255.255.0, also written **/24**, the first 24 bits are the network and the last 8 are the host.

ADDRESS 192.168.10.25 /24	VALUE
Network address	192.168.10.0
First usable host	192.168.10.1
Last usable host	192.168.10.254
	192.168.10.255
Usable hosts	254

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Two devices with the same network part talk directly. Anything else goes to the **default gateway**, the router's address on this network.

Ranges worth knowing

RANGE	MEANING
10.0.0.0/8	Private
172.16.0.0/12	Private (172.16.x.x to 172.31.x.x)
192.168.0.0/16	Private
127.0.0.0/8	Loopback: 127.0.0.1 is "this machine"
169.254.0.0/16	Link-local: the device gave up waiting for DHCP
100.64.0.0/10	Carrier-grade NAT, used by some ISPs

Private addresses are free to use inside any network but are never routed on the internet; NAT translates them on the way out.

Checking a machine

WINDOWS CMD

```
ipconfig /all
```

SHELL

```
ip addr show
ip route
```

You are looking for four things: the address, the mask, the default gateway and the DNS servers. If any of them is missing or wrong, very little else will work.

rong

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- **169.254.x.x:** DHCP failed. Check the cable, the VLAN and the DHCP server.

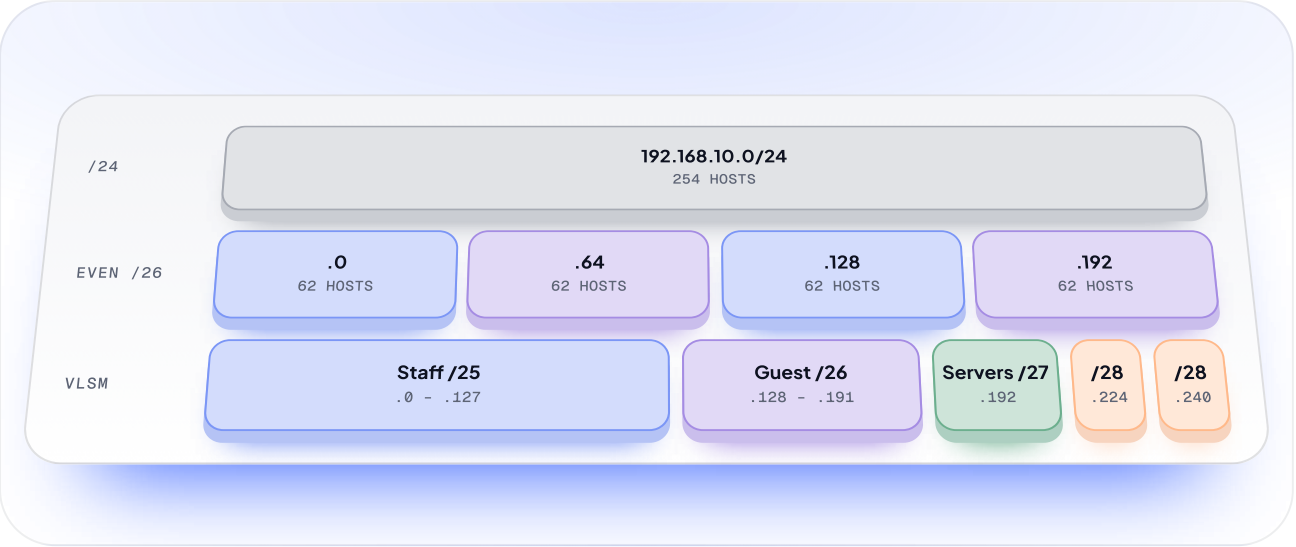
- **Right address, no gateway:** the device can reach its own subnet but nothing beyond.
- **Duplicate address warning:** two devices were given the same static address, or a static address sits inside the DHCP pool.
- **Wrong mask:** the device thinks a local neighbour is remote (or the reverse) and sends traffic the wrong way.

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CHAPTER 2 · SUBNETTING / VLSM

Subnetting and VLSM

Cutting one network into right-sized pieces using the block size.



One /24 split evenly into four /26 blocks, then split by need with VLSM: big blocks first.

Subnetting cuts one network into smaller ones. You do it to keep departments apart, to limit broadcasts and to stop wasting addresses. The trick that makes it quick is the **block size**.

The block size

The block size is 256 minus the interesting octet of the mask, the one that is neither 255 nor 0.

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
SKIP TO CONTENT	55.255.240	16	14

PREFIX	MASK	BLOCK SIZE	USABLE HOSTS
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Subnets start at multiples of the block size. For /26 that is .0, .64, .128 and .192.

Which subnet is this address in?

Take 192.168.10.150/26. The block size is 64. The multiples of 64 are 0, 64, 128, 192. 150 falls between 128 and 192, so:

- Network: 192.168.10.128
- Broadcast: 192.168.10.191 (one less than the next block)
- Hosts: .129 to .190

VLSM: sizes that fit

Variable Length Subnet Masking means each subnet gets the size it needs instead of all being equal. Say you have 192.168.10.0/24 and need:

NEED	HOSTS	PREFIX	SUBNET
Staff	100	/25	192.168.10.0/25
Guest Wi-Fi	50	/26	192.168.10.128/26
Servers	20	/27	192.168.10.192/27
Printers	10	/28	192.168.10.224/28
Spare		/28	192.168.10.240/28

TIP

Always allocate the **largest** subnet first. Each block then starts on a clean ing overlaps.

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A quick check before you deploy

- Does every subnet start on a multiple of its block size?
- Does each one end before the next one begins?
- Is there room to grow? A subnet that is 95% full today will be a problem next year.

When it goes wrong

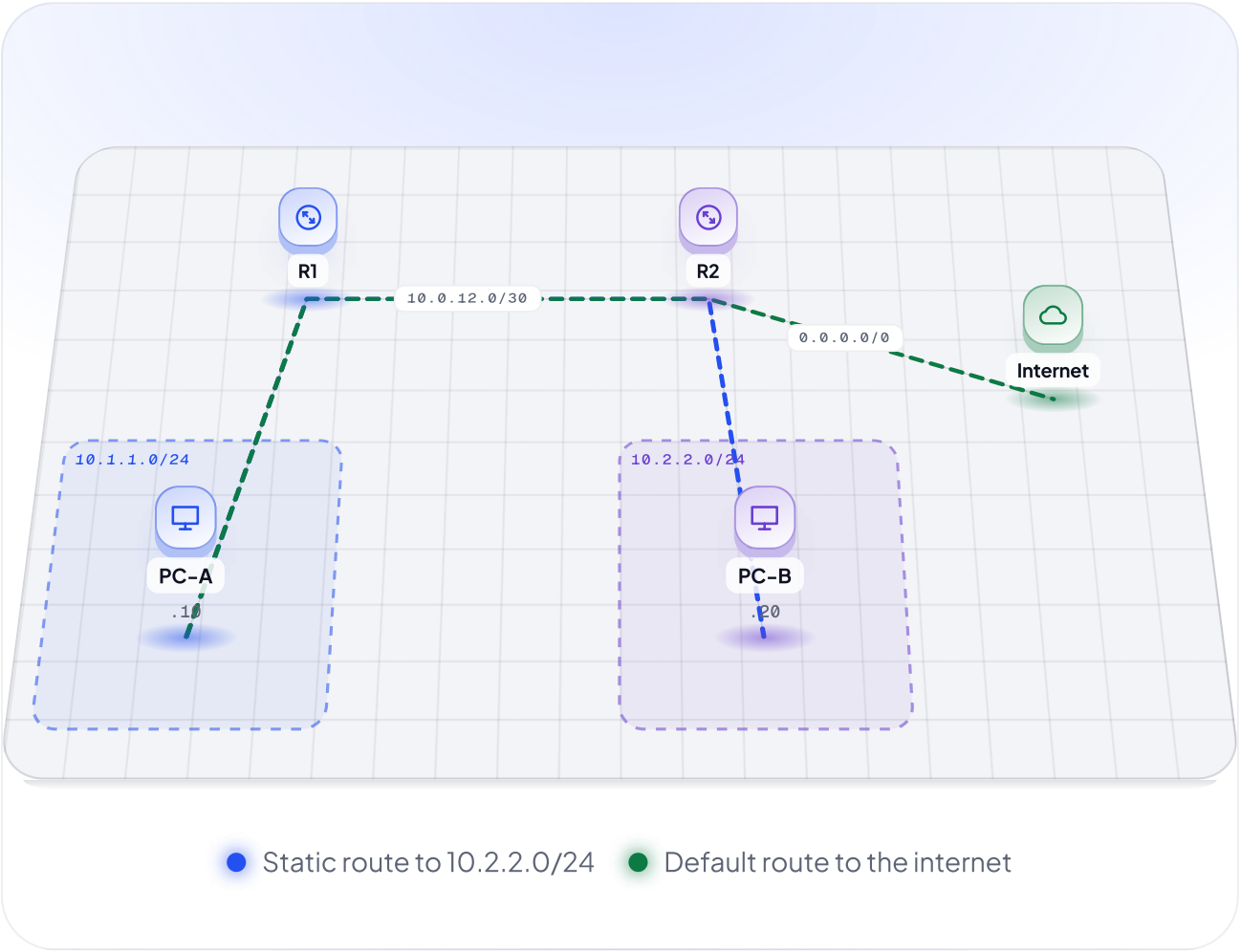
- **Overlapping subnets:** two routers both claim the same addresses and traffic goes to the wrong one.
- **A host on the network or broadcast address:** it cannot be used. Pick an address inside the host range.
- **Mask typed wrong on one device:** it disagrees with its neighbours about who is local.

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CHAPTER 3 · STATIC ROUTING

Static Routing

Telling a router by hand where a remote network is, and the default route for everything else.



Each router knows its own LAN. A static route on each side points at the other, and R2 sends the rest to the internet.

A router only knows the networks it is directly connected to. To reach anything further away it needs a route. A **static route** is one you type in yourself: "to reach network X, send the packet to that next router".

Reading a routing table

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```text
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
C    10.0.12.0/30 is directly connected, GigabitEthernet0/1
S    10.2.2.0/24 [1/0] via 10.0.12.2
S*   0.0.0.0/0 [1/0] via 10.0.12.2
```

`C` is connected, `S` is static and `*` marks the default route. The numbers in brackets are the administrative distance and the metric.

Adding routes

On R1, send traffic for R2's LAN to R2:

```cisco ios
ip route 10.2.2.0 255.255.255.0 10.0.12.2
```

On R2, the return route to R1's LAN, and a default route to the internet:

```cisco ios
ip route 10.1.1.0 255.255.255.0 10.0.12.1
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

> **WARNING**
>
> Every route needs a **return route**. If R1 knows how to reach 10.2.2.0 but R2 does not know how to get back to 10.1.1.0, pings leave and never come home. It looks exactly like a firewall problem.

Longest match wins

When several routes match, the router picks the most specific one. A packet for 10.2.2.50 matches both 10.2.2.0/24 and 0.0.0.0/0; the /24 wins. The default route only catches what nothing else matches.

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Floating static routes

Give a backup route a higher administrative distance, and it is only used when the main route disappears:

```
CISCO IOS

ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip route 0.0.0.0 0.0.0.0 198.51.100.1 200
```

When to use static routes

- Small networks with one or two routers.
- A default route to the ISP.
- Stub networks with only one way in and out.

Once you have more than a handful of routers, or more than one path, a dynamic protocol such as OSPF is easier and recovers from failures by itself.

When it goes wrong

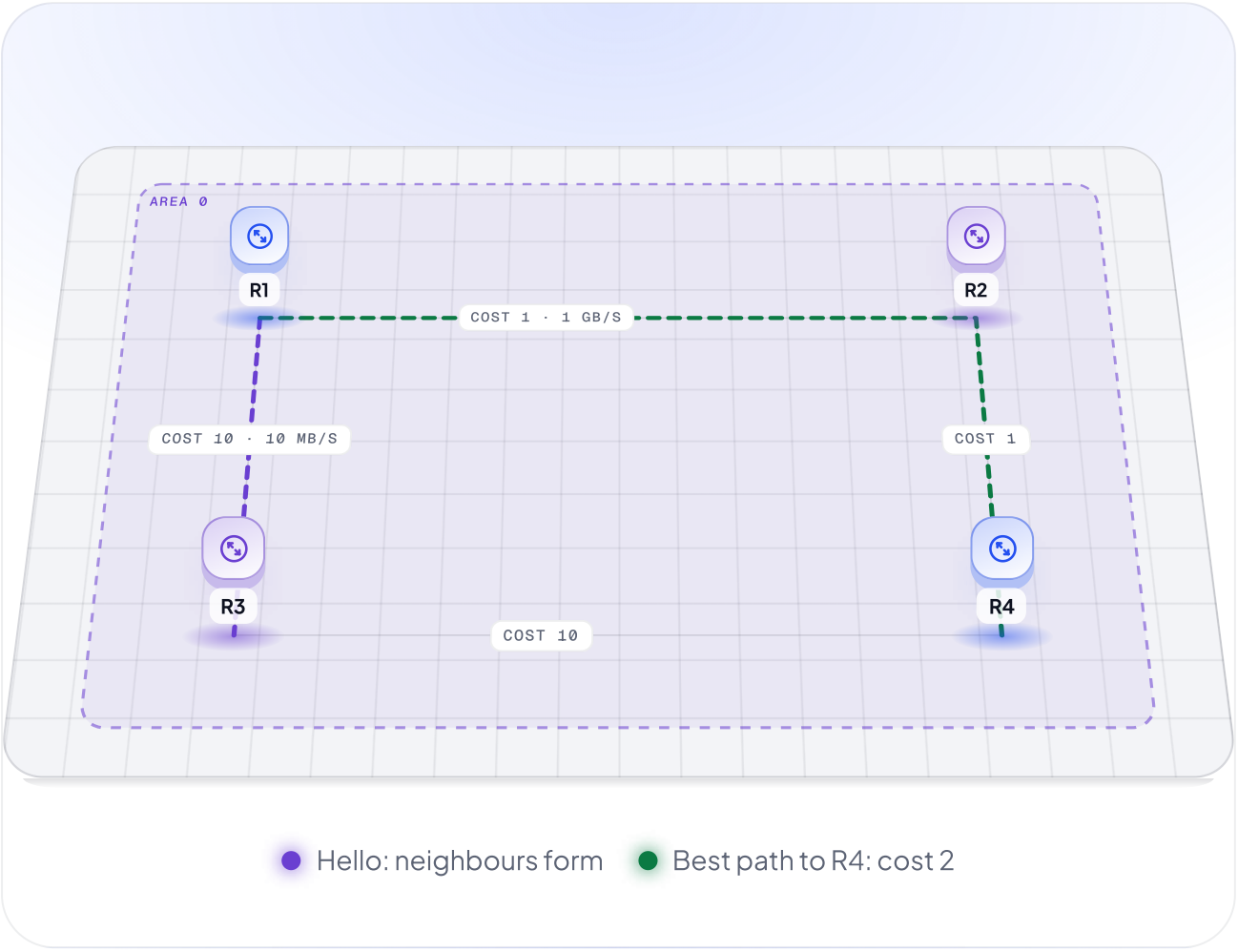
- **Ping one way only:** missing return route.
- **Route in the config but not in the table:** the next hop is unreachable or its interface is down.
- **Traffic goes to the internet instead of a branch:** the specific route is missing, so the default route caught it.

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CHAPTER 4 · OSPF

OSPF

Routers that find each other, share a map of links and pick the lowest-cost path.



Every router builds the same map and picks the cheapest path. R1 reaches R4 over two fast links, cost 2, not the slow pair at cost 20.

OSPF (Open Shortest Path First) lets routers find their neighbours, share a map of every link in the network and work out the best path themselves. When a link fails, they recalculate within seconds. Nobody has to edit a route.

How it works

the same link send Hello packets every 10 seconds and become
ne settings match.

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2. **Adjacency:** neighbours exchange link-state advertisements (LSAs) until every router has the same database, the map.
3. **SPF:** each router runs Dijkstra's shortest-path algorithm on that map, with itself at the centre.
4. **Routes:** the best paths go into the routing table, marked `O`.

Cost

OSPF's metric is **cost**, worked out from bandwidth: reference bandwidth divided by interface bandwidth. The default reference is 100 Mb/s, which makes 100 Mb/s, 1 Gb/s and 10 Gb/s all cost 1. Raise the reference on every router so fast links are told apart:

CISCO IOS

```
router ospf 1
 auto-cost reference-bandwidth 10000
```

The path with the lowest total cost wins, even if it has more hops.

A basic configuration

CISCO IOS

```
router ospf 1
 router-id 1.1.1.1
 network 10.0.12.0 0.0.0.3 area 0
 network 10.1.1.0 0.0.0.255 area 0
 passive-interface GigabitEthernet0/0
```

- The `network` statement uses a **wildcard mask**, the inverse of the subnet mask.
- `passive-interface` advertises the LAN without sending Hellos to the PCs on it.
- Small networks need only **area 0**. Larger ones add more areas, all joined to area 0.

Checking it

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```
show ip ospf neighbor
show ip route ospf
```

```
show ip ospf interface brief
```

A healthy neighbour shows the state **FULL** (or `2WAY` between two non-designated routers on a shared segment).

Neighbours that will not form

Both sides must agree on:

- Subnet and mask on the shared link
- Area number
- Hello and dead timers
- Authentication, if used
- MTU (otherwise they stick in `EXSTART`)

And each router needs a unique router ID.

> TIP
>
> If a neighbour never appears, check `show ip ospf interface` on both ends first. Nine times in ten the area or the timers do not match.

When it goes wrong

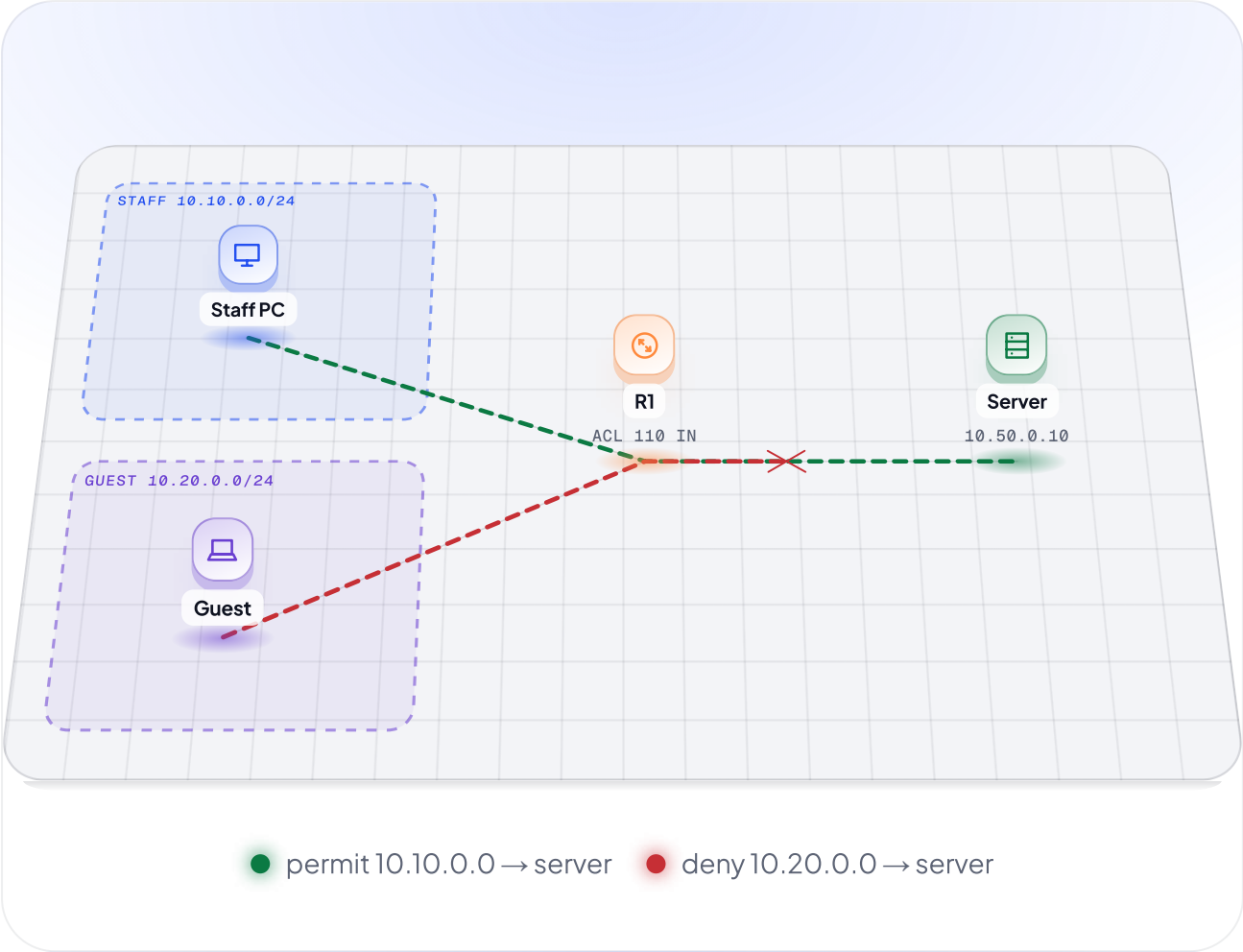
- **Neighbour stuck in INIT:** Hellos only work in one direction, often an ACL or a one-way link.
- **Route missing:** the network was never advertised, or the interface is passive on the wrong side.
- **Traffic over a slow link:** reference bandwidth is still the default, so the costs look equal.

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CHAPTER 5 · ACLS

Access Control Lists

Permit and deny rules on a router interface, read top to bottom.



An ACL on R1 lets staff reach the server and stops guests. The first matching line wins; the last line is an unseen deny.

An access control list is an ordered list of permit and deny rules that a router checks against each packet. It is the simplest way to say "guests may use the internet but not the server".

The three rules of ACLs

1. **Top to bottom:** the router reads the lines in order.
2. **First match wins:** as soon as a line matches, the router acts and stops reading.
3. **Implicit deny:** every ACL ends with an invisible `deny any`. A packet that matches nothing is dropped.

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So put specific lines first, general lines last, and remember to permit what should get through.

Standard and extended

- **Standard** ACLs match only the source address. Place them close to the destination.
- **Extended** ACLs match source, destination, protocol and port. Place them close to the source, so unwanted traffic is dropped early.

Example: keep guests off the server

Staff are 10.10.0.0/24, guests 10.20.0.0/24, and the server is 10.50.0.10.

CISCO IOS

```
ip access-list extended GUEST-RULES
 remark guests may not reach the server
 deny   ip 10.20.0.0 0.0.0.255 host 10.50.0.10
 remark everything else from guests is allowed
 permit ip 10.20.0.0 0.0.0.255 any
!
interface GigabitEthernet0/1.20
 ip access-group GUEST-RULES in
```

Notes:

- The `0.0.0.255` is a wildcard mask: 0 means "must match", 255 means "any value".
- `host 10.50.0.10` is short for `10.50.0.10 0.0.0.0`.
- `in` checks packets as they arrive on the interface; `out` checks them as they leave.

Checking it

CISCO IOS

```
show access-lists
```

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ch counter. If a line you expect to match stays at zero, the traffic is not arriving where you think, or an earlier line is catching it.

WARNING

Never apply a new ACL to the interface you are connected through without a permit for your own session. Test on a console, or use `reload in 5` as a safety net.

When it goes wrong

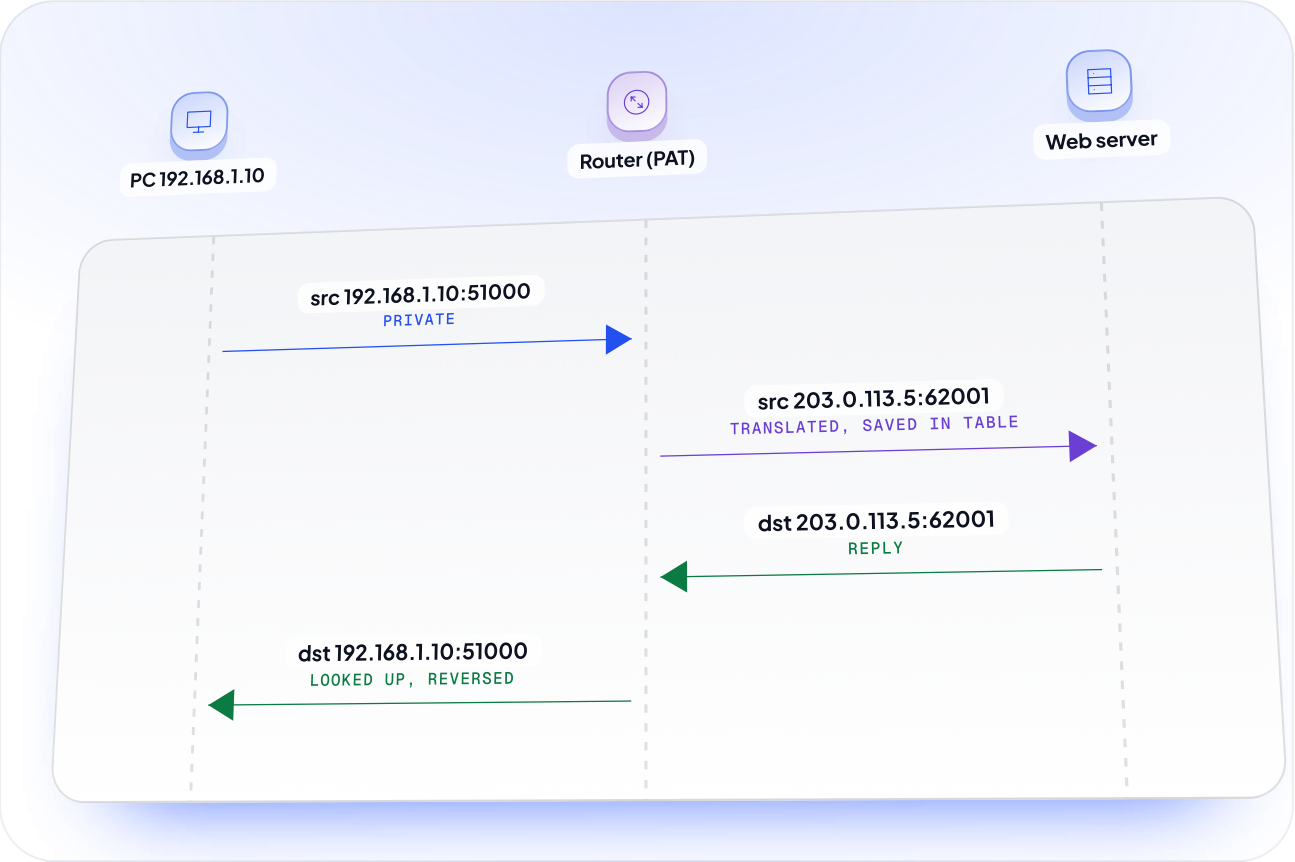
- **Everything stopped after applying the ACL:** there is no permit line, so the implicit deny blocked it all.
- **The deny does nothing:** a broader permit above it matches first.
- **Return traffic is blocked:** ACLs are not stateful. Allow replies with `established` for TCP, or use a firewall.

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CHAPTER 6 · NAT/PAT

NAT and PAT

Sharing one public address between many private devices by rewriting addresses and ports.



The router swaps the private source for its public address and a free port, remembers the pair, and reverses it on the reply.

Private addresses such as 192.168.1.10 cannot travel on the internet. **NAT** (Network Address Translation) rewrites them to a public address as packets leave, and back again as replies arrive. **PAT** (Port Address Translation), also called NAT overload, goes further and lets hundreds of devices share a single public address by also changing the source port.

What the router does

1. A PC sends a packet from 192.168.1.10, port 51000.

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the source with its public address and a free port: 203.0.113.5,

3. It writes the pair into its **translation table**.
4. The reply comes back to 203.0.113.5:62001. The router looks it up and sends it to 192.168.1.10:51000.

The port is what tells two PCs' conversations apart, which is why one address is enough.

Types of NAT

TYPE	MAPPING	TYPICAL USE
Static NAT	one private ↔ one public, permanent	a server that must be reachable
Dynamic NAT	private → one from a pool of publics	rare today
PAT	many private → one public + ports	every home and office router

Configuring PAT

CISCO IOS

```
interface GigabitEthernet0/0
 ip nat inside
interface GigabitEthernet0/1
 ip nat outside
!
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

Port forwarding

To let the internet reach a server inside, map a public port to it with static NAT:

CISCO IOS

```
ip nat inside source static tcp 192.168.1.20 443 203.0.113.5 443
```

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Checking it

```cisco-ios
show ip nat translations
show ip nat statistics
```

Each line shows the inside local, inside global, outside local and outside global addresses. If translations appear but replies do not arrive, the problem is after the router.

> **NOTE**
>
> NAT hides inside addresses, but it is not a firewall. Treat any port you forward as exposed to the whole internet.

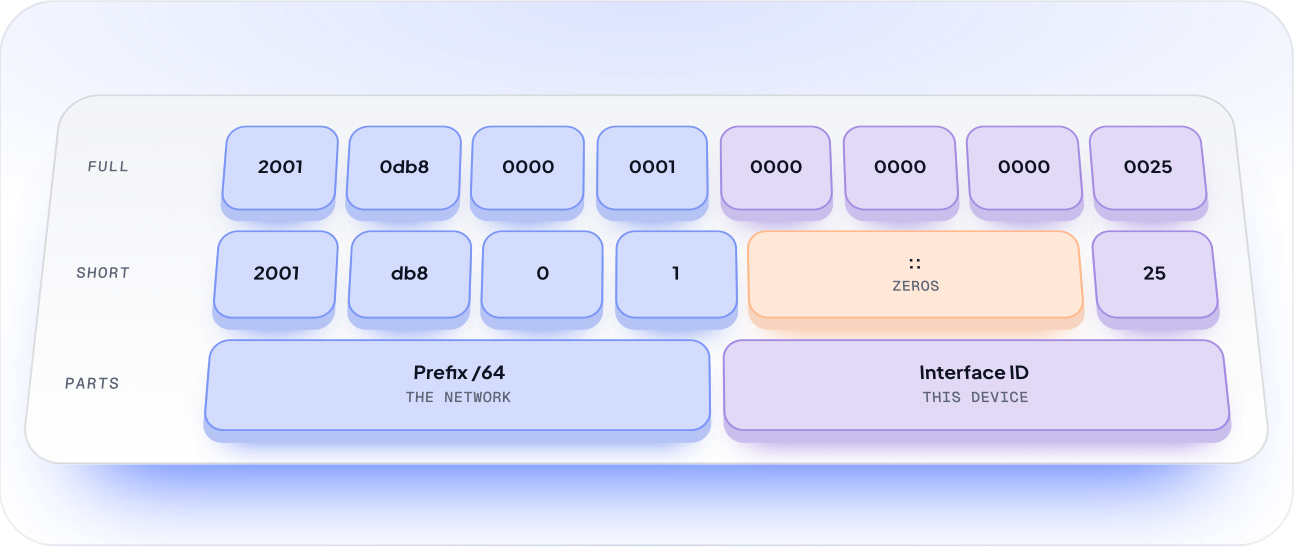
When it goes wrong

- **Internet works for nobody:** inside and outside are swapped on the interfaces, or the ACL does not match the LAN.
- **Port forward does not work from inside the office:** many routers do not "hairpin". Test from outside, or use internal DNS.
- **Games or VoIP struggle:** some protocols carry addresses inside the payload and need helpers or a different setup.

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IPv6 Basics

Reading and shortening 128-bit addresses, and the prefix and interface parts.



Eight groups of four hex digits. Drop leading zeros, replace one run of zero groups with ::, and split at /64.

IPv4 has about 4.3 billion addresses, and they ran out years ago. IPv6 uses 128 bits instead of 32, which is enough for every device to have a real, globally unique address with no NAT in the way.

Reading an address

An IPv6 address is eight groups of four hexadecimal digits, separated by colons:

```text
2001:0db8:0000:0001:0000:0000:0000:0025
```

Two rules make it shorter:

1. **Drop leading zeros** in any group: `0db8` becomes `db8`, `0001` becomes `1`.
2. **Replace one run of all-zero groups** with `::`. Only once per address, or it would

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```text
```

```
2001:db8:0:1::25
```

Prefix and interface ID

Almost every LAN uses a **/64**. The first 64 bits are the network prefix, the same for every device on the link. The last 64 bits are the interface ID, unique to each device.

PART	EXAMPLE
Prefix	2001:db8:0:1::/64
Interface ID	::25

Address types to recognise

STARTS WITH	TYPE	MEANING
2000::/3	Global unicast	Public, routable on the internet
fe80::/10	Link-local	Every interface has one; never leaves the link
fc00::/7	Unique local	Private, like 10.x.x.x
ff00::/8	Multicast	IPv6 has no broadcast; multicast replaces it
::1	Loopback	This machine

Getting an address

- **SLAAC:** the router advertises the prefix and the device builds its own interface ID. No DHCP server needed.
- **DHCPv6:** a server hands out addresses, like DHCP for IPv4.
- **Static:** typed in, as for servers and routers.

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Checking a machine

WINDOWS CMD

```
ipconfig
ping -6 google.com
```

CISCO IOS

```
show ipv6 interface brief
show ipv6 route
```

TIP

A device that shows only an `fe80::` address has IPv6 switched on but never heard a router advertisement. Check that the router has `ipv6 unicast-routing` enabled.

When it goes wrong

- **Some sites slow, others fine:** IPv6 is partly broken, so the browser tries it, waits and falls back to IPv4.
- **No global address:** no router advertisements reaching the device.
- **Firewall rules only for IPv4:** IPv6 traffic is either wide open or blocked. Write rules for both.

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CHAPTER 8 · PING / TRACEROUTE

ping and traceroute

Testing reachability, and finding the hop where a path breaks.



traceroute sends packets with TTL 1, 2, 3… Each router that drops one to zero replies, so every hop shows itself.

`ping` asks "can I reach you?" `traceroute` asks "which way does the path go, and where does it stop?" Between them they solve a large share of network tickets.

ping

ping sends ICMP **echo request** packets and waits for **echo replies**.

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```
ping 8.8.8.8
ping -n 20 192.168.1.1
ping -t 10.0.0.5
```

TEXT

```
Reply from 8.8.8.8: bytes=32 time=18ms TTL=117
```

RESULT	WHAT IT SUGGESTS
Replies, low time	The path works
Request timed out	No reply: down, filtered or no route back
Destination host unreachable	A router or your own PC has no way to deliver it
General failure	Your own network stack or adapter is the problem
Some replies lost	Congestion, a bad cable or Wi-Fi trouble

NOTE

Many servers and firewalls block ping on purpose. A failed ping proves less than a successful one.

Ping in a sensible order

1. `127.0.0.1`: the network stack works.
2. Your own IP address: the adapter is up.
3. The default gateway: the local network works.
4. A public IP such as `8.8.8.8`: routing and the internet link work.
5. A name such as `google.com`: DNS works.

The first step that fails tells you where to look.

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Every IP packet carries a **TTL** (time to live). Each router lowers it by one and drops the packet when it reaches zero, sending back a "time exceeded" message. traceroute uses this on purpose: it sends packets with TTL 1, then 2, then 3, so each router on the path reveals itself in turn.

WINDOWS CMD

```
tracert google.com
```

SHELL

```
traceroute google.com
mtr google.com
```

Reading the output

TEXT

```
 1    1 ms    1 ms    1 ms  192.168.1.1
 2    9 ms    8 ms    9 ms  10.20.0.1
 3    *       *       *     Request timed out.
 4   18 ms   17 ms   18 ms  142.250.183.14
```

- Each line is a hop, with three timing samples.
- `* * *` on one hop, with later hops answering, just means that router does not reply to traceroute. Ignore it.
- `* * *` from some hop **to the end** means the path stops there.
- A sudden jump in time that continues to every later hop marks a slow link. A jump on one hop only is that router being slow to answer, not slow to forward.

When it goes wrong

- **Gateway pings, internet does not:** the router, its NAT or the ISP link.
- **IP works, names do not:** DNS, not the network.
- **Trace loops between two addresses:** a routing loop, usually two static routes pointing at each other.